

Scrutinize

Multi-Modal AI Ingestion, Retrieval & Agentic Search

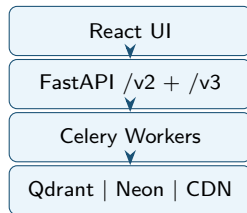
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AI News Project

July 15, 2026

What is Scrutinize?

- Unified platform for **text, audio, and video** ingestion
- Natural-language Q&A via **hybrid RAG + live web search**
- Persistent **project workspaces** and chat history (V3)
- Agentic pipeline with routing, synthesis, and quality evaluation



Four Layers

Client (React) → FastAPI → Celery Workers → Qdrant + Neon + Cloudinary

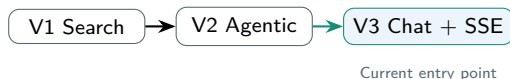
Evolution: Old → New Implementation

Legacy (V1 / early V2)

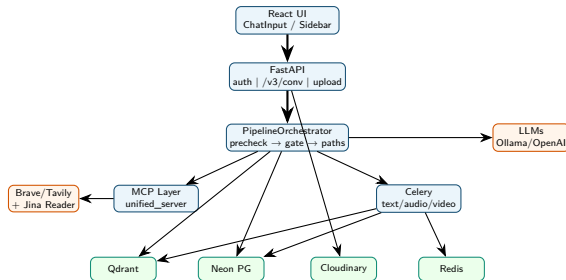
- Single POST /v2/search endpoint
- Non-streaming responses
- Gate-only routing (no precheck)
- Dense-only vector search
- No persistent conversations
- Password auth only

Current Stack

- V3 SSE streaming chat API
- Retrieval precheck (fast-path routing)
- Hybrid dense + BM25 (RRF fusion)
- MCP tool layer (web search, PDF)
- Google OAuth + JWT sessions
- Project-scoped workspaces + API keys



High-Level System Architecture

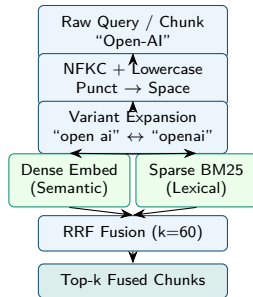


Improvement: Spelling & Keyword Normalization

Keyword Search Normalization

(keyword_search_utils.py)

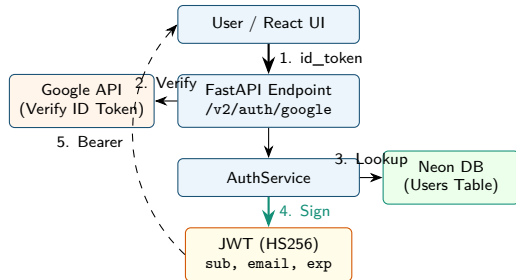
- **Unicode NFKC Normalization:** Standardizes character representations and lowercases all inputs.
- **Punctuation to Spaces:** Replaces hyphens (-) and underscores (_) with spaces to unify compound words.
- **Compound Collapsing:** Generates collapsed and expanded variants (e.g., open ai ↔ openai) for high recall.
- **Filename Stem Extraction:** Extracts clean, searchable stems from paths and URLs.
- **Index Enrichment:** Combines raw content, title, and filename variants to build a rich sparse index.



Feature: Google OAuth & JWT Authentication

Authentication & Session Flow

- **Frontend Login:** User signs in via Google OAuth in the React UI and receives a secure `id_token`.
- **Backend Verification:** The token is sent to `POST /v2/auth/google`. FastAPI verifies the token signature and audience with Google's API.
- **User Provisioning:** `AuthService` checks Neon Postgres. If the email is new, a user record is created on the fly.
- **JWT Generation:** A signed HS256 JWT is generated containing user claims (`sub`, `email`, `iat`, `exp`).
- **Access Control:** The client attaches the JWT as a Bearer token in the Authorization



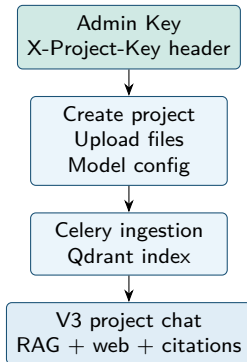
Feature: Project-Wise Management & API Key Access

Project CRUD (Neon Postgres)

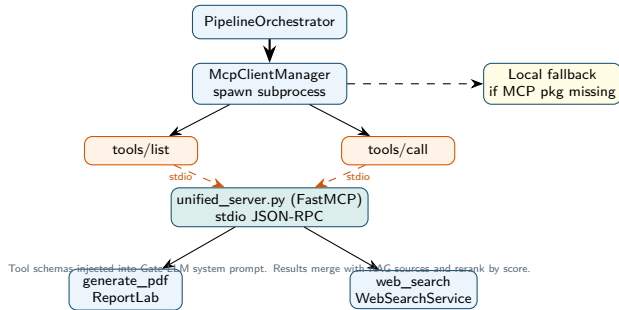
- Create projects with admin & client API keys
- Upload files scoped to `project_id`
- Per-project model overrides (gate, rewriter, decision)
- `project_members` for team access

Two conversation scopes (V3)

- **general** — web-only, no project RAG
- **project** — full hybrid pipeline + indexed files

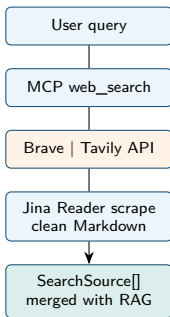


Feature: MCP Tool Layer Architecture

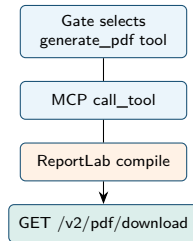


Features: Web Search (Tavily) & PDF Generation

Web Search Pipeline

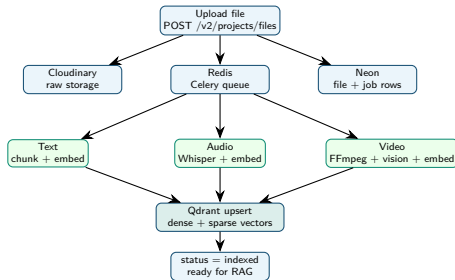


PDF Generation



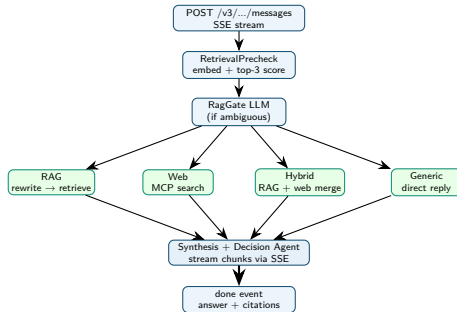
web_search_mode toggle: auto | always | never

Feature: Real-Time File Indexing Pipeline



Async Celery workers process uploads in background; Fly.io scaler wakes workers on demand.

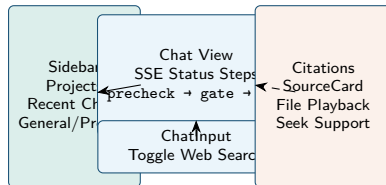
Query Pipeline: Four Routing Paths



Improvement: UI Enhancements

Unified Chat & Workspace UI

- **ConversationChatView**: SSE stream with live status steps (precheck, gate, rewrite, retrieval, synthesis, evaluation).
- **Sidebar**: Project workspaces, recent chats, and scope selection (general vs project).
- **ChatInput**: Three-state web search mode toggle (auto, always, never).
- **SourceCard**: Clickable citations with direct file playback and seek support.
- **UploadView / LibraryView**: Multi-modal file management (text, audio, video).



Future Improvements

Planned Enhancements

- **Knowledge graph** context management
- **LangSmith** — LLM tracing & eval
- **Flowchart MCP server** — diagram generation tool
- **DocLinQ** integration
- Native multi-vector points (CLIP visual, CLAP audio)

Current: Hybrid RAG + MCP

+ Knowledge Graph

+ LangSmith Observability

+ Multi-modal Vectors

Try Scrutinize

Demo Flow

- 1 Sign in with **Google OAuth**
- 2 Create a **project workspace** and upload documents
- 3 Start a **project-scoped chat** — ask about your files
- 4 Toggle **web search mode** and compare RAG vs hybrid answers
- 5 Request a **PDF report** from the assistant
- 6 Inspect **citations** and source cards

Thank You

“The best way to predict the future is to invent it.”

— Alan Kay

Ammer Saeed

Scrutinize — Multi-Modal AI Ingestion & Retrieval

GitHub | FastAPI + React + Qdrant + MCP